

MUHAMMAD HAIDER ZAIDI

☎ +49 1523 1698988 ✉ mhaiderzaidi21@gmail.com in linkedin.com/in/muhammad-haider-zaidi
🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++

Tools/Frameworks: PyTorch, TensorFlow, Docker, FastAPI, AWS SageMaker, Kubernetes, TensorRT, Git, Airflow

Competence: Machine Learning, Deep Learning, MLOps, Computer Vision, Perception, Segmentation, Time Series Prediction, Trajectory Prediction, Data Engineering, LLM, Edge Computing, Agile Development, Transformers

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **deep learning models** for **metrological applications**, improving model performance and training efficiency.
- Explored **novel architectures** and physics-informed hybrid loss functions to improve model accuracy and stability.
- Co-authored a publication with PhD researchers for the FLOMEKO conference, demonstrating a **scientific background** and developed dashboards for data visualization and **model performance monitoring**.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo **segmentation accuracy** to 94% and increased inference speed by 5x using **AI algorithms**, deployed with **Docker** and **FastAPI** for real-world impact.
- Developed a **real-time** tennis foul **detection system** leveraging **TensorRT**, reducing compute cost and time by 6x, demonstrating capabilities for **edge processing**.
- Managed **MLOps pipelines** on **AWS SageMaker**, addressing **YOLO model data drift**, and maintained **Kubernetes infrastructure** for **continuous integration** and **agile development**.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed an **AI chatbot application** using **OpenAI API** and relevant **LLM tools/frameworks**; contributed to full-stack development of an **AI-based services website** with **Python/Flask**.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built **web scrapers** and **ETL pipelines** for structured/unstructured data, automating **data ingestion** and **cleaning workflows** for client analytics use cases.

Projects

CAFA 6: Protein Function Prediction 2022

Deep Learning Competition

- Architected a **weighted-probability ensemble** integrating DPFunc/DeepGOPlus, leveraging AlphaFold for **feature extraction**, achieving a 4% increase in model accuracy across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting 2022

Thesis Project

- Developed a **thesis project** focusing on **real-time Bitcoin price forecasting** using **modern ML techniques** for **time series prediction**, demonstrating strong analytical and scientific skills.

Saaz: AI Self-Playing Chitralli Sitar 2023

Signal Processing, Music Generation

- Collaborated on the **world's first AI-driven self-playing Chitralli Sitar**, integrating **signal processing** and **music generation LLMs**.
- Project featured on Coke Studio in a music video, showcasing real-world application and innovative problem-solving in a public domain.

Hugging Face Open-Source

Contributor

- Co-authored **Zero-shot Image Classification technical documentation** and implementation page on the Hugging Face Hub, demonstrating expertise in **perception**, open-source contribution, and collaboration.

Education

Friedrich Alexander Universität Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Pursuing advanced studies in **Artificial Intelligence** with a focus on **Machine Learning**, **Deep Learning**, and their applications in complex systems.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Awarded **Dean's List of Honor** for academic excellence, recognizing achievement within the top 5% of the class.
- Completed comprehensive coursework in **Computer Science**, building a robust foundation for advanced AI/ML studies.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python